PINN design: architecture and inductive biases

Module 4 ▪ two networks share one physics loss ▪ six design choices enforce SEIRD structure exactly

The observability problem

- Surveillance reports two series: cumulative cases C and deaths D
- Hidden compartments S, E, I are **never measured**
- $\beta(t)$ and μ must be inferred from C and D alone

Observable re-parameterisation

Set $C = I + R + D$ (everything symptomatic). Then $S = N - E - C$ exactly, so mass conservation *cannot be violated* for any network output.

Dual-network architecture

- **State net** $g_\theta(\tau)$: 3×48 tanh units, 4 outputs

$$e = e_0\, e^{\tau g_1},\ i = i_0\, e^{\tau g_2},\ c = c_0\, e^{\tau g_3},\ d = d_0 + \tau g_4$$

- **Transmission net** $h_\phi(\tau)$: 3×32 tanh units, $\beta(\tau) = \text{Softplus}(h_\phi(\tau) + b) > 0$

Six inductive biases

1. **Hard ICs:** exponential form $e_0 e^{\tau g_1}$ satisfies $e(0) = e_0$ exactly; τ factor means the gradient of the IC loss is always zero
2. **Positivity:** Softplus and exponential forms keep every state > 0
3. **CFR bounds:** $\mu = \gamma\,\text{CFR}/(1 - \text{CFR})$, $\text{CFR} \in (0.2, 0.8)$, blocks the degenerate $\mu \to 0$ solution
4. **Fourier features:** $\sin(k\pi\tau')$ and $\cos(k\pi\tau')$, $k = 1, \ldots, 4$ overcome spectral bias so $\beta(t)$ changes can be represented inside the window
5. **Relative ODE residuals:** each equation divided by its state, so the early (small) phase costs as much as the late (large) phase
6. **Smooth prior on β:** penalty on $(d\beta/d\tau)^2$ makes the inverse problem well-posed

What the architecture guarantees without training: $S + E + I + R + D = N$ to machine precision; $e, i, c > 0$; initial conditions exact; $\text{CFR} \in (0.2, 0.8)$.

Loss function and training protocol

Module 4 ▪ four terms; Adam brings it close, L-BFGS closes the gap

Total loss

$$\mathcal{L} = \underbrace{\mathcal{L}_C + \mathcal{L}_D}_{\text{data}} + \mathcal{L}_{\text{ode}} + w_{\text{sm}}\,\mathcal{L}_{\text{smooth}}$$

$$\mathcal{L}_C = \tfrac{1}{n_C}\textstyle\sum_k \big(c(\tau_k) - C_k/K\big)^2$$

$$\mathcal{L}_D = \tfrac{1}{n_D}\textstyle\sum_k \big(d(\tau_k) - D_k/K\big)^2$$

$$\mathcal{L}_{\text{ode}} = \tfrac{1}{N_f}\textstyle\sum_j \Big[\Big(\frac{e' - Tf_e}{e}\Big)^2 + \Big(\frac{i' - Tf_i}{i}\Big)^2 + \cdots\Big]_{\tau_j}$$

$$\mathcal{L}_{\text{smooth}} = \tfrac{1}{N_f}\textstyle\sum_j \big(\frac{d\beta}{d\tau}(\tau_j)\big)^2$$

K = max training case count; $N_f = 200$ collocation points; $w_{\text{sm}} = 10^{-4}$ (synthetic), 10^{-3} (real data); $w_{\text{data}} = w_{\text{ode}} = 1$.

Why relative ODE residuals?

An epidemic grows by orders of magnitude. With plain (absolute) residuals the early phase costs almost nothing. The E6 ablation shows absolute residuals give 75–128% forward error vs. below 2×10^{-3} with relative residuals.

Two-stage optimiser

- **Stage 1 – Adam** (LR 2×10^{-3}, cosine decay, 2000 iters): fresh random collocation points each iteration; fast escape from saddle points
- **Stage 2 – L-BFGS** (strong-Wolfe line search, 2500–6000 iters, fixed linspace grid): second-order convergence to a tight solution
- All weights in **double precision** (float64) for accurate physics residuals
- **Ensemble:** 5 independent seeds; median is the point estimate

Forecasting

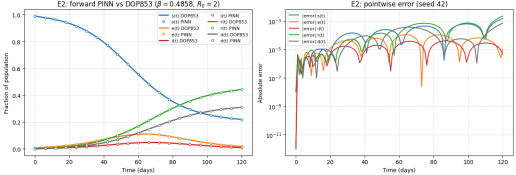
Hold β at its mean over the last 14 training days; keep learned μ; integrate forward with DOP853. Forecast is the median over seeds. The assumption that β stays constant limits predictive horizon.

E6 ablation headline

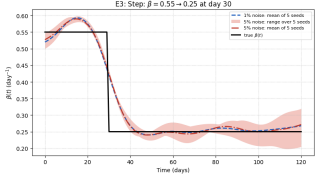
Adam-only: $e(t)$ error 96%; $w_{\text{ode}} = 10$: errors triple; no Fourier features: β RMSE +38%.

Synthetic experiments: E2 forward, E3 inverse $\beta(t)$, E4 hidden states

Module 4 ▪ no real data; ground truth is known; PINN is tested against DOP853



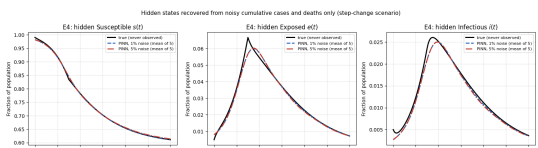
E2: forward PINN (markers) vs. DOP853 (lines) and pointwise error



E3: recovered $\beta(t)$ (band = seed range, 5% noise)

Quantity	1% noise	5% noise
$\beta(t)$ rel. L_2	$11.0\% \pm 0.7\%$	$11.6\% \pm 0.6\%$
μ rel. error	$4.6\% \pm 0.3\%$	$5.4\% \pm 1.0\%$
hidden $s(t)$	$0.36\% \pm 0.03\%$	$0.46\% \pm 0.09\%$
hidden $e(t)$	$5.8\% \pm 0.6\%$	$6.4\% \pm 0.7\%$
hidden $i(t)$	$5.8\% \pm 0.4\%$	$6.5\% \pm 0.8\%$

E3/E4: mean $\pm$ SD over 5 seeds; β drops 0.55$\to$0.25 at day 30



E4: reconstructed $s(t)$, $e(t)$, $i(t)$ from cases and deaths only

E2: worst compartment $< 2 \times 10^{-3}$; conservation 2×10^{-16} (machine zero). **E3:** a sharp β change recovered within 11%; 5$\times$ more noise costs < 1 pp. **E4:** susceptible fraction recovered within 0.5% from *zero* direct measurements.